

INVITED SPEAKER TALK

Topology-Preserving Contrastive Learning for Taxonomy Induction in Thai Lexical Semantics



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About the speaker

Peerachet Porkaew received his B.Eng. degree in Computer Engineering from Chiang Mai University in 2007 and his M.S. degree in Computer Science from the University of Chinese Academy of Sciences in 2018. Since joining NECTEC in 2008, he has been an active researcher in the Artificial Intelligence Research Group, where he has conducted research in natural language processing and artificial intelligence.

His research interests include Machine Translation, Large Language Models (LLMs), Deep Learning, and Natural Language Processing (NLP), with a particular focus on developing and applying AI technologies for language understanding and generation.

Abstract

Lexical relation classification—the task of identifying semantic links between words such as Synonyms and Hypernyms—is a foundational requirement for constructing and traversing taxonomic Knowledge Graphs (KGs). Many general-purpose semantic representation objectives emphasize symmetric similarity, whereas lexical entailment requires explicit directional structure. In standard dense retrievers, the tendency of symmetric similarity objectives to treat directional lexical relations as bidirectionally interchangeable causes a systematic spatial collapse we formalize as the “Symmetric Trap”, severely corrupting hierarchical graph traversal and downstream KG construction.

To address this, we propose LCCL (Linguistically-Constrained Contrastive Loss), a topology-aware framework introducing a relation-aware contrastive objective that selectively applies symmetric attraction to symmetric relations and directional norm constraints (L_{asym}) to asymmetric IS-A relations directly in Euclidean space. Built upon an independently cached Siamese encoder with a standard pairwise interaction head, LCCL avoids numerical instability from explicit hyperbolic projection. We benchmark our framework on a Thai WordNet taxonomy benchmark comprising 23,600 pairs across 7 relation types. While LCCL matches the classification performance of the strongest Cross-Encoder baseline (Macro F1 = 0.676 ± 0.008 , McNemar’s $p = 0.841$), it decisively rescues the underlying hierarchical geometry, verified by Kolmogorov-Smirnov separation tests (KS = 0.496 vs. 0.412 for SimCSE).

CHAIRPERSON

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